



Course Syllabus

General Course Information

Identifying Course Details

MATH 106-01 (CRN: 7309)

Course Title

Finite Mathematics

Term

Fall 2026, full semester

Format

Traditional, MWF 10:00-10:50, SH 205

Catalog Description

This course is designed to provide students with opportunities to think critically and analytically, to formulate and communicate informed, reasoned opinions, to develop their abilities to find logical approaches to solve mathematical problems, and to reflect upon the processes of problem solving. The topics treated in this course help students work with quantitative information and make critical decisions of the type encountered in both personal life and in academic disciplines.

Credit Hours

3

Prerequisites

A student must satisfy one of the following:

- ACT Math score of 19 or higher
- SAT Math score of 510 or higher
- Concurrently enrolled in MATH 106S

Course Objectives

Upon completing this course, students will be able to ...

1. Apply different types of reasoning and logical connectors to analyze arguments, think critically, and develop rationals.
2. Use elementary set theory concepts to solve problems related to surveys, arguments, and probabilities.
3. Create and interpret statistical graphs, including frequency distributions and measures of center, to analyze data. Use mean and standard deviation to fit data to a normal distribution and estimate population percentages.
4. Identify and apply counting methods (e.g., permutations, combinations) and probability rules to solve problems. Use dimensional analysis to convert between units within the US system, the metric system, and between the two systems. Apply linear and exponential models to solve financial problems.
5. Use intermediate algebra concepts (such as expressions, equations, and inequalities), to solve and verify quantitative problems.

General Education Competencies

1. **Analytical Reasoning:** Evaluating, solving, and visually representing real world problems using quantitative and/or qualitative data; mastering scientific concepts and reasoning.
2. **Life-Long Learning:** Becoming independent learners and problem solvers, accepting responsibility for their personal and professional welfare and growth.

Instructor Information

Instructor Name and Title

Pai Song, Associate Professor of Mathematics

Office Location and Office Hours

Science Hall 202A, MW: 11AM-1PM, F: 11AM-12PM

Contact Information

(304)462-6156

pai.song@glennville.edu

Expected Response Time

Usually within 24 hours, except on holidays and weekends.

Other Pertinent Information

Personal website address: <http://www.paisong.org>

Course-Specific Information and Requirements

Course Materials

Required Materials

- OpenStax. (2023). *Contemporary Mathematics*. OpenStax. ISBN-13: 978-1-951693-68-8
This text can be accessed online at no cost to the student here:
<https://openstax.org/details/books/contemporary-mathematics>

Technical Requirements

A laptop or tablet.

Course Content and Structure

This course is designed to provide students with opportunities to think critically and analytically, to formulate and communicate informed, reasoned opinions, and to develop their abilities to find logical approaches to solve mathematical problems and reflect upon the processes of problem solving. The topics treated in this course help students work with quantitative information and make critical decisions of the type encountered in both personal life and in academic disciplines. The course is organized into four modules, each building on core topics in liberal arts mathematics — set theory and logic, algebraic reasoning, counting and probability, and statistics. Each module is broken into lessons that pair a reading assignment with practice problems, and students will have weekly quizzes, periodic module exams, and a cumulative final exam review to reinforce and assess their understanding. A general outline of the topics and their sequence is below; the exact weekly schedule with dates will be provided separately each semester, as it may shift from year to year.

Course Outline by Module

Module 1 – Sets, Logic, and Reasoning

- Lesson 1: Sets and Subsets
- Lesson 2: Venn Diagrams and Set Operations
- Lesson 3: Surveys with Venn Diagrams
- Lesson 4: Statements, Negations, and Quantified Statements
- Lesson 5: Compound Statements and Connectives
- Lesson 6: Arguments and Euler Diagrams
- Lesson 7: Inductive and Deductive Reasoning
- Exam 1

Module 2 – Numbers and Algebraic Reasoning

- Lesson 1: Rational Numbers
- Lesson 2: Real Numbers and Their Properties
- Lesson 3: Dimensional Analysis
- Lesson 4: Linear Equations in One Variable and Proportions
- Lesson 5: Applications of Linear Equations

- Lesson 6: Linear Inequalities in One Variable
- Lesson 7: Simple and Compound Interest
- Exam 2

Module 3 – Counting and Probability

- Lesson 1: Fundamental Counting Principle and Permutations
- Lesson 2: Combinations
- Lesson 3: Fundamentals of Probability
- Lesson 4: Probability with the Fundamental Counting Principle, Permutations, and Combinations
- Lesson 5: Events Involving Not and Or; Odds
- Lesson 6: Events Involving And
- Exam 3

Module 4 – Statistics

- Lesson 1: Sampling, Frequency Distributions, and Graphs
- Lesson 2: Measures of Central Tendency
- Lesson 3: Measures of Dispersion
- Lesson 4: The Normal Distribution
- Exam 4

Final Exam Review

Quizzes are given weekly (typically on Fridays) covering the most recent lessons, and the schedule accounts for university holidays and breaks (e.g., Labor Day, Fall Recess, Thanksgiving Break) as they fall during the semester. Any changes to the schedule or assignments will be communicated to students by the instructor.

Weekly Schedule

Full Semester Weekly Schedule

Week	Dates	Course Assignments	Notable GSU Info
1	8/17/26 – 8/21/26	Mod 1 Lesson 1 – Sets and Subsets (1.1, 1.2) Mod 1 Lesson 2 – Venn Diagrams and Set Operations (1.3, 1.4, 1.5) Quiz 1 (Friday) -- Syllabus and Module 1 Lesson 1	8/21/26: Last day to drop classes without a W on the transcript.
2	8/24/26 – 8/28/26	Mod 1 Lesson 3 – Surveys with Venn Diagrams (1.3) Mod 1 Lesson 4 – Statements, Negations, and Quantified Statements (2.1) Quiz 2 (Friday) -- Module 1 Lessons 2, 3	N/A
3	8/31/26 – 9/4/26	Mod 1 Lesson 5 – Compound Statements and Connectives (2.2) Mod 1 Lesson 6 – Arguments and Euler diagram (2.7) Quiz 3 (Friday) -- Module 1 Lesson 4,5	N/A

Week	Dates	Course Assignments	Notable GSU Info
4	9/7/26 – 9/11/26	Mod 1 Lesson 7 – Inductive and Deductive Reasoning (1.1) Exam 1 – Friday, September 11	9/7/26: Labor Day. No classes; University closed.
5	9/14/26 – 9/18/26	Mod 2 Lesson 1 – Rational Numbers (3.4) Mod 2 Lesson 2 – Real Numbers and Their Properties (3.6) Quiz 4 (Friday) -- Module 2 Lesson 1, 2	N/A
6	9/21/26 – 9/25/26	Mod 2 Lesson 3 – Dimensional Analysis (9.1 - 9.5) Mod 2 Lesson 4 – Linear Equations in One Variable and Proportions (5.2) -- Note: There are two homework assignments for this lesson. Quiz 5 (Friday) - Module 2 Lessons 3, 4	N/A
7	9/28/26 – 10/2/26	Mod 2 Lesson 5 – Applications of Linear Equations (5.1, 5.2) Mod 2 Lesson 6 – Linear Inequalities in One variable (5.3) Quiz 6 (Wednesday) -- Module 2 Lessons 5, 6	N/A
8	10/5/26 – 10/9/26	Mod 2 Lesson 7 – Simple and Compound Interest (6.3, 6.4) Exam 2 – Wednesday, October 7	Midterm Week 10/9/26: Fall Recess. No classes; University open.
9	10/12/26 – 10/16/26	Mod 3 Lesson 1 – Fundamental Counting Principle and Permutations (7.1, 7.2) Quiz 7 (Friday) -- Module 3 Lesson 1	
10	10/19/26 – 10/23/26	Mod 3 Lesson 2 – Combinations (7.3) Mod 3 Lesson 3 – Fundamentals of Probability (7.5) Quiz 8 (Friday) -- Module 3 Lessons 2, 3	N/A
11	10/26/26 – 10/30/26	Mod 3 Lesson 4 – Probability with the Fundamental Counting Principle, Permutations, and Combinations (7.6) Mod 3 Lesson 5 – Events Involving Not and Or; Odds (11.6) Quiz 9 (Friday) -- Module 3 Lessons 4, 5	10/23/26: Last day to drop classes with a W on the transcript.
12	11/2/26 – 11/6/26	Mod 3 Lesson 6 – Events Involving And (11.7) Exam 3 – Wednesday, November 5	N/A
13	11/9/26 – 11/13/26	Mod 4 Lesson 1 – Sampling, Frequency distributions, and Graphs (8.1, 8.2) Mod 4 Lesson 2 – Measures of Central Tendency (8.3) Mod 4 Lesson 3 – Measures of Dispersion (8.4, 8.5) Quiz 10 (Friday) -- Module 4 Lessons 1,2	N/A

Week	Dates	Course Assignments	Notable GSU Info
14	11/16/26 – 11/20/26	Mod 4 Lesson 4 – The Normal Distribution (8.6) Exam 4 – Friday, November 21	N/A
N/A	11/23/26 – 11/27/26	None	Thanksgiving Break. No classes. University open Monday – Wednesday.
15	11/30/26 – 12/4/26	Final Exam Review	N/A
16	12/7/26 – 12/10/26	Final Exam, the final exam will be given on Thursday from 8:00 to 10:00 Link to GSU final exam schedule: LINK	Finals Week

Methods of Evaluation

How Evaluations Impact Course Grade

Method(s) of Evaluation	Percentage or Points Possible
Attendance	10%
Homework	20%
Quizzes	10%
Exams	40%
Final Exam	20%
Total	100%

Grading Scale

The following grading scale applies to all graded work in this course, as well as your overall course grade.

Grading Scale

Letter Grade	Percentage
A	90–100
B	80–89
C	70–79
D	60–69
F	0–59

Class Attendance Policy

Attendance is required for all classes. To comply with federal guidelines, students are expected to be present at all class sessions to be eligible for financial aid through Title IV of the Higher Education Act of 1965. It may be necessary for the student to be absent from scheduled classes or laboratories for personal reasons. On such occasions, all matters related to a student's absence are to be arranged between the student and the instructor, including making up missed work. The student is responsible for the academic consequences of any absences.

Occasionally, students may be absent from scheduled classes to participate in officially sanctioned university activities (known as institutional absences). Instructors are obligated to respect institutional absences and may not penalize a student for such an absence if the student makes up any graded work missed due to the absence in a timely manner as determined by the instructor. An institutional absence does not change deadlines for submitting assignments, but the instructor will allow students to make up exams, quizzes, presentations, or any other course requirements that have an impact on the course grade conducted during the missed class(es). Students are responsible for all coursework missed due to absences and must initiate any request to make up coursework promptly as stipulated by the instructor on their syllabi. The instructor is free to assign alternate, equivalent work for any in-class work a student missed due to an institutional absence. Instructor accommodations may be made prior to or after the institutional absences at the discretion of the instructor. A list of students who are excused during a specific time period will be maintained by the Office of Academic Affairs and circulated to instructors.

In addition, instructors may excuse absences due to extenuating circumstances beyond student control as if they were institutional absences. However, if an instructor chooses to excuse such absences, they must excuse them consistently for all students. Furthermore, at the instructor's

discretion, an instructor may require the student to provide documentation of the circumstances leading to the absences and approve or deny the request for an excused absence based on said documentation.

Students and instructors are obligated to adhere to the stated requirements of each course. An instructor's attendance policy may not contradict institutional policy.

Course Policies on Late Work and Make-Up Work

Students are responsible for completing all work by the posted due dates.

- **Homework and Quiz:** Homework and quizzes may be submitted late with a **5% penalty per day late**, as long as they remain available in Knewton.
- **Exams and Final Exam:** Module exams and the final exam may not be completed after the due date. Make-up exams will only be considered for serious, documented circumstances. Students must contact the instructor before the deadline whenever possible, or as soon as possible in an emergency.
- Documentation may be required. Late or make-up work is not guaranteed and will be handled on a case-by-case basis.

Academic Integrity

Students are expected to complete all coursework honestly and independently unless collaboration is explicitly permitted. Cheating on quizzes or exams, plagiarism, unauthorized assistance on homework assignments, and disruptive classroom behavior will not be tolerated and will be handled in accordance with the university's Oath of Academic Excellence and academic dishonesty policy.

The use of artificial intelligence (AI) in each Glenville State University course is governed by the individual course syllabus. Use of AI in this course outside acceptable AI use as defined by the course syllabus will constitute academic dishonesty. Faculty use of tools and methods to detect violations of academic integrity, including but not limited to AI detection tools, is expressly permitted by Glenville State University policy.

Level of AI Use Allowed: BY PERMISSION ONLY

Students must obtain permission from the course instructor prior to using Automated Writing Tools (including but not limited to ChatGPT and other AI tools) for any assignment in this course. The use of AI tools in this course without the instructor's permission violates the university policy on academic dishonesty and will be handled in accordance with that policy.

Technological Information

Brightspace (GSU's Learning Management System)

At a minimum, all GSU courses should use the LMS to provide access to the course syllabus and the course grade book. If you need assistance accessing or navigating Brightspace, please contact the GSU Help Desk at 304-462-6145 between 8:00 a.m. and 4:00 p.m. Eastern (GMT – 4:00) on regular U.S. workdays, when the university is open. You may also [email the Brightspace Help Desk \(Brightspace.Help@glenville.edu\)](mailto:Brightspace.Help@glenville.edu), and someone will reply as soon as possible during normal business hours.

Other Campus Systems

If you need support with university-provided software or systems, such as email, passwords, or EdNet, please call the Office of Technology for support at 304-462-6140 between 8:00 a.m. and 4:00 p.m. Eastern (GMT – 4:00) on regular U.S. workdays, when the university is open.

University-Wide Policies

All university policies are in effect during this course for relevant policy and procedures in the classroom.

ADA Students Statement

It is the policy of Glenville State University to provide reasonable accommodations for qualified individuals with documented disabilities. This University will adhere to all applicable federal, state, and local laws, regulations, and guidelines regarding providing reasonable accommodations for affording equal educational opportunities. It is the student's responsibility to provide documentation of a disability to Pioneer Support, located on the third floor of the Robert F. Kidd Library. You may also [email Pioneer Support \(Pioneer.Support@glenville.edu\)](mailto:Pioneer.Support@glenville.edu). Staff will assist students and faculty in arranging appropriate accommodations. This is in accordance with Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act of 1990.

Other Pertinent Information

Course Evaluations

Course evaluations will be available at the end of the course. Student feedback is important; as such, all students are expected to provide their candid and constructive feedback for all courses taken.

Reserve Clause

The instructor reserves the right to revise, alter, or amend this syllabus as necessary. Students will be notified in writing / e-mail of any such changes. Note also that the duties and obligations of Glenville State University may be modified or suspended immediately and without notice

because of causes beyond Glenville's reasonable control and occurring without its fault or negligence including, but not limited to, acts of God, fire, war, governmental action, terrorism, epidemic, pandemic, weather, national emergencies, or other threats to the safety of students, faculty, or staff. Such events may require the university to alter the academic year schedule or provide alternate means of instruction, including distance or remote learning.

Glenville State University Mission

Glenville State University prepares and inspires students to be thoughtful, productive, engaged, and responsible citizens who contribute to the well-being of their community, state, nation, and world.